

What are Occupational Standards (OS)?

- OS describe what individuals need to do, know and understand in order to carry out a particular job role or function
- OS are performance standards that individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding



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Introduction

Qualifications Pack- Mobile Equipment Operator

SECTOR: Iron & Steel

SUB-SECTOR: Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory

REFERENCE ID: ISC/Q0401

ALIGNED TO: NCO-2004/NIL

Title of Job: The job involves inspection of mobile equipment like Fork Lift truck, Skid-loader/Bob Cat, Pay Loader and JCB, as used in Steel Plant for proper working condition, loading & driving the vehicle for carrying materials from one place to another inside the plant. This job also involves carrying out basic maintenance of the equipment.

Personal Attributes: The job requires the candidate to concentrate on the job at hand and complete it without any accidents. Diligence, presence of mind and hardworking are the desired traits for individuals in this role. Understanding the need to take initiative and follow instructions provided by supervisor.



Job Details

Qualifications Pack Code	ISC/Q0401		
Job Role	Mobile Equipment Operator		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron & Steel	Drafted on	17/09/2014
Sub-sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory	Last reviewed on	30/12/2014
Occupation	Iron Making	Next review date	30/12/2015
NSQC Clearance on	18/06/2015		



Job Role	Mobile Equipment Operator
Role Description	The job involves inspection (according to a checklist) of mobile equipment like Fork Lift, Skid-loader/Bob Cat, Pay Loader and JCB as used in Steel Plant for proper working condition, loading & driving the vehicle for carrying materials from one place to another. This also involves carrying out basic maintenance of the equipment.
NSQF level	3
Minimum Educational Qualifications	Class X pass Valid Driving License mandatory
Maximum Educational Qualifications	ITI pass Valid Driving License mandatory
Training (Suggested but not mandatory)	• Technical and gallery/hand-held training • Refresher training if absent from works for a period of one year or more before re-employment • Must possess licence for heavy vehicle driving • Basic maintenance and troubleshooting • 2 weeks hands on training in steel plant (mandatory) • Safety practices
Minimum Job Entry Age	18 years
Experience	• Minimum 1 year of loading, driving & unloading experience with the vehicles • In lieu of minimum qualification the incumbent should have minimum 3-4 years of experience of driving heavy vehicles
Occupational Standards (OS)	Compulsory: ISC/N0401: Prepare the vehicle ISC/N0402: Perform mobile equipment operations ISC/N0403: Perform basic maintenance and troubleshooting of the mobile equipments ISC/N0008: Use basic health & safety practices at work place ISC/N0009: Works effectively with others Optional: N/A
Performance Criteria	As described in the relevant NOS units



Definitions

Keywords /Terms	Description
Core Skills/Generic Skills	Core Skills or Generic Skills are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.
Function	Function is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organization.
Knowledge and Understanding	Knowledge and Understanding are statements which together specify the technical, generic, professional and organizational specific knowledge that an individual needs in order to perform to the required standard.
National Occupational Standards (NOS)	NOS are Occupational Standards which apply uniquely in the Indian context
Occupation	Occupation is a set of job roles, which perform similar/related set of functions in an industry.
Organisational Context	Organisational Context includes the way the organization is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Performance Criteria	Performance Criteria are statements that together specify the standard of performance required when carrying out a task.
Qualifications Pack(QP)	Qualifications Pack comprises the set of NOS, together with the educational, training and other criteria required to perform a job role. A Qualifications Pack is assigned a unique qualification pack code.
Qualifications Pack Code	Qualifications Pack Code is a unique reference code that identifies a qualifications pack.
Scope	Scope is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.
Sector	Sector is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.



Sub-Sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Sub-functions	Sub-functions are sub-activities essential to fulfil the achieving the objectives of the function.
Technical Knowledge	Technical Knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Unit Code	Unit Code is a unique identifier for a NOS unit, which can be denoted with an 'N'
Unit Title	Unit Title gives a clear overall statement about what the incumbent should be able to do.
Vertical	Vertical may exist within a sub-sector representing different domain areas or the client industries served by the industry.
Keywords /Terms	Description
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
OEM	Original Equipment Manufacturer
OS	Occupational Standard(s)
QP	Qualifications Pack
5 S	Technique of maintaining orderliness –Japanese terminology
CP	Control Plan
WI	Work Instructions



ISC/N0401: Prepare the vehicle



Overview

This unit is about preparing the mobile equipment (vehicle) for activities that need to be carried out.



Unit Code	ISC/N0401
Unit Title (Task)	Prepare the vehicle
Description	This unit is about inspecting & preparing the Vehicle (mobile equipment) for activities that need to be carried out.
Scope	This OS unit/task covers the following: - Understand the job requirements according to the checklist - Conduct pre-operation checks according to a checklist and to ensure that the vehicle is working & is safe to use
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Understand the job requirements according to the checklist	To be competent, the user/individual on the job must be able to: PC1. Interpret and understand the job requirements PC2. Prepare a work plan to carry out the job
Conduct pre-operation checks according to a check list and to ensure the Vehicle is safe to use	To be competent, the user/individual on the job must be able to: PC3. Check oil levels of engine, transmission, radiator coolant and brake PC4. Check differential and hydraulic oil levels PC5. Check the hydraulic hose and ram for leakages PC6. Check condition of parking brake, main horn, reverse horn, and head light PC7. Check the various controls, gauges, warning lamp and other safety devices PC8. Check fan belt tension, electrolyte level and terminal tightness PC9. Check and adjust driving position, rear and side mirrors, seat belts PC10. Clean air filter dust bowls PC11. Drain water and sediment from the fuel tank PC12. Top up coolant and oil in engine, transmission, etc. if necessary PC13. Check electrolyte level, terminals, seating arrangements of the battery PC14. Apply grease to all greasing pins and pivot points PC15. Ensure the locking bar is in position to prevent the front and rear chassis moving and creating a crushing zone (articulated machines only) PC16. Keep footplates and steps clean and free from mud, dirt and oil. PC17. Report defects precisely to the supervisor if beyond scope of his role
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Job specific documents e.g. daily maintenance checklist & its importance KA2. Safety policy of the company as applicable to his role KA3. Emergency plan of the work site KA4. Risk and impact of not following defined procedures/work instructions

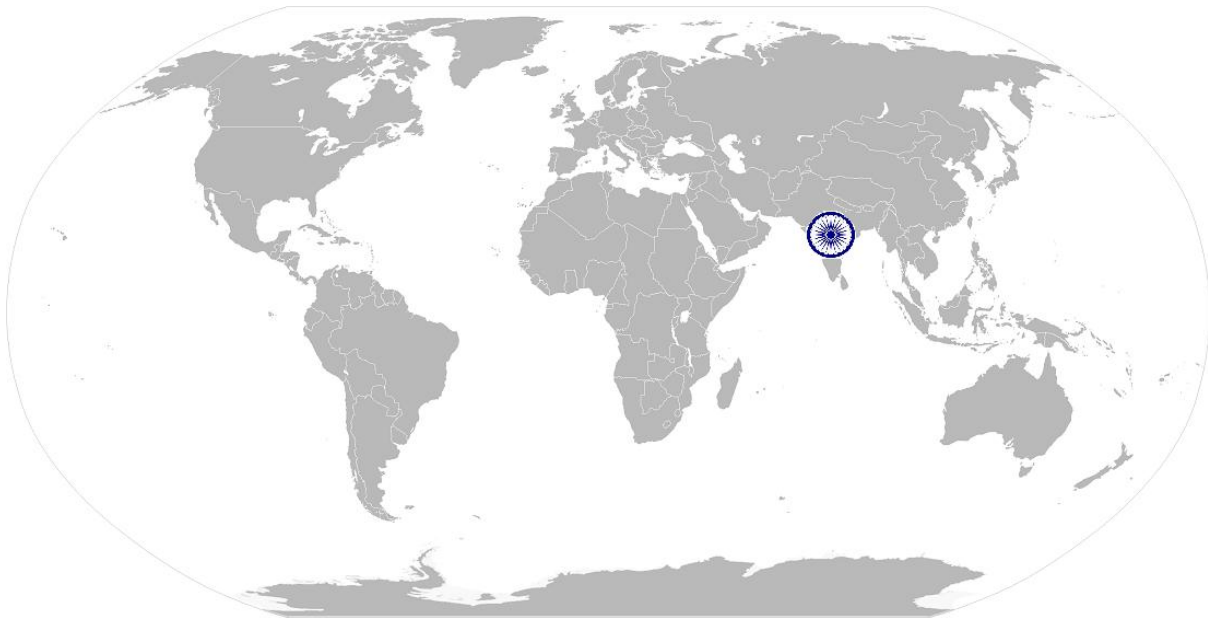


	KA5. Escalation matrix for reporting identified problems
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Basic functions and operation of the vehicle (Fork lift, JCB, Skid Loader/Bob cat, Pay loader) & their main sub-parts like parking & service brakes, boom, forks, basket, bucket etc.) KB2. Loading techniques under different conditions KB3. Basic driving rules and techniques, including front & reverse movement and associated safety rules
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/Generic Skills	Oral Communication (Listening and Speaking skills) The user/individual on the job needs to know and understand how to: SA1. Effectively explain to supervisors, operators and helpers about equipment operation, process steps and other operational requirements, in English/Hindi/Local language (E/H/LL) SA2. Answer the queries raised by the operative team as well as concerned departments
B. Professional Skills	Time Management The user/individual on the job needs to know and understand how to: SB1. Prioritize and execute tasks within scheduled time limit without breaking any statutory norms. Fault Diagnostic and Maintenance The user/individual on the job needs to know and understand how to: SB2. Check for damage in mobile equipment components (e.g. forks, chain, battery, motors, hydraulic ram, bucket, brakes, horn, lights etc.) and report it for rectification to the appropriate authority, if beyond his ability



NOS Version Control

NOS Code	ISC/N0401		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and steel	Drafted on	18/09/2014
Industry Sub-sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory	Last reviewed on	30/12/2014
Occupation	Iron Making	Next review date	30/12/2015





ISC/N0402: Perform mobile equipment operations



Overview

This unit is about performing operations on the mobile equipment to load, carry and transfer the material as required by the engaging department.



Unit Code	ISC/N0402
Unit Title (Task)	Perform mobile equipment operations
Description	This unit is about performing operations on the vehicle (mobile equipment) to transfer the material as required by the engaging department.
Scope	This OS unit/task covers the following: - Start & operate the vehicle to lift the specific load (coils, slag, debris, bricks, slabs, zinc/lead/aluminium ingots, spares, equipment etc.) to be transferred - Drive the vehicle safely on the approved route - Discharge the load in correct & safe way at the desired location - Park the vehicle with the arm/fork/boom/basket/bucket in lowered condition
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Start & operate the vehicle to lift the specific load (coils, slag, debris, bricks, slabs, zinc/lead/aluminium ingots, spares, equipment etc.) to be transferred	To be competent, the user/individual on the job must be able to: PC1. Start the engine using the push-button switch and key switch together PC2. Use the priming pump and pre-heater to start the engine in cold weather conditions PC3. Load the material to be transferred safely & correctly, ensuring that it is not damaged or spilt or does not slip/fall off during loading or during journey
Drive the vehicle safely on the approved route	To be competent, the user/individual on the job must be able to: PC4. Always wear safety belt & helmet while driving PC5. Ensure proper functioning of rear view mirror and audio visual alarm PC6. Not use mobile phones while driving PC7. Look out for pedestrians, people working, stationary objects and hazards such as trenches, potholes, cables, caring for overhead pipes, power lines, confined space etc. PC8. Operate the shifting quadrant device for selecting the correct gear setting PC9. Ensure vehicle load and operating speed are within limits specified by the manufacturer & do not cross the limits specified in the company PC10. Adhere to time limits given by supervisor PC11. Follow the road signs and locally developed signals PC12. Keep a safe distance from a tip edge and use an approved stop block before tipping over an edge PC13. Select & use the right type of brake in different situations during journey



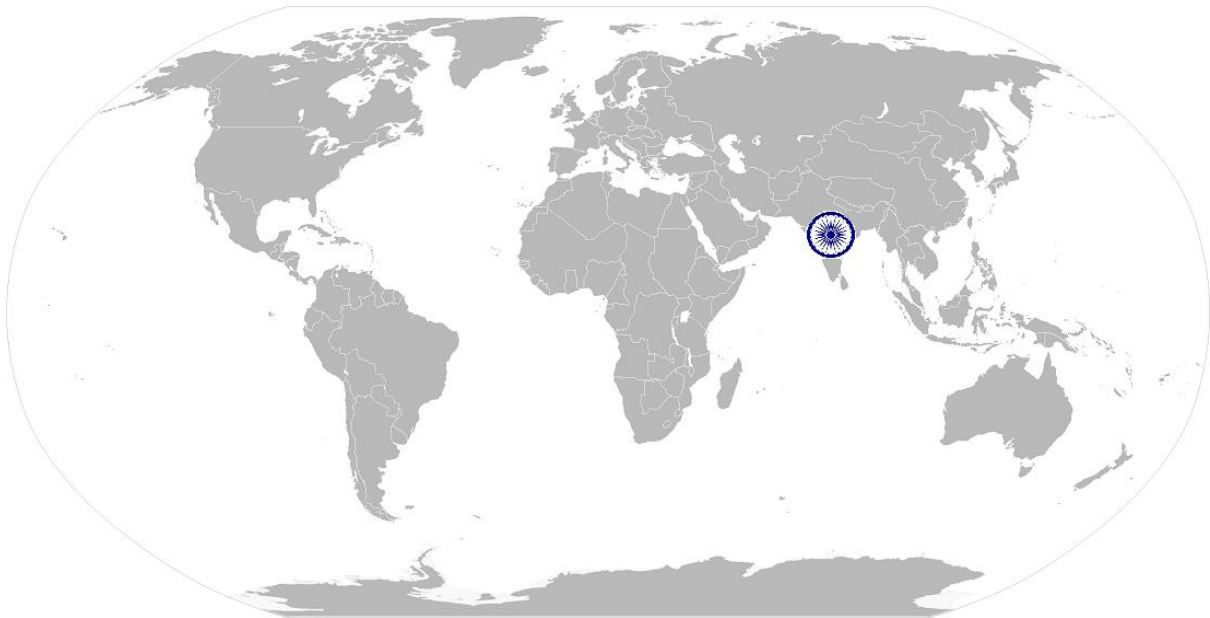
Discharge the load in correct & safe way at the desired location	To be competent, the user/individual on the job must be able to: PC14. Discharge the load safely at the position and in the manner designated by the supervisor PC15. Ensure that vehicle is always parked on firm, level ground; with handbrake applied and drive & controls disengaged
Park the vehicle with the arm/fork/boom/basket/bucket in lowered condition	To be competent, the user/individual on the job must be able to: PC16. Put the front accessory (boom, fork, basket/bucket) in lowered condition touching the ground; switch the engine off, apply handbrake and alight from the seat PC17. Ensure that nobody except the authorised driver ever boards, travels on or stands near the Vehicle
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Job specific documents e.g. daily maintenance checklist and importance of the same KA2. Safety policy of the company as applicable to his role KA3. Emergency plan of the specific work site KA4. Risk and impact of not following the defined procedures/work instructions KA5. Escalation matrix for reporting identified problems KA6. Implications of delays in process to the company
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Service brakes and parking brakes KB2. Instrument panel, their location and operation KB3. Controls, levers and switches in order to operate the mobile equipment KB4. General safety rules vis-à-vis loading, hauling, stocking and parking KB5. Operation and safety aspects of the equipment, especially of the forks, boom, basket, bucket etc.
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing skills and reading Skills The user/ individual on the job needs to know and understand how to: SA1. Read & understand all displays and be aware of all standard safety symbols SA2. Construct simple sentence & write technical observations & requisitions in English/Hindi/Local Language Oral Communication (Listening and Speaking skills) The user/individual on the job needs to know and understand how to:



	SA3. Communicate with supervisors/juniors/colleagues/casual workers in English/Hindi/Local Language SA4. Respond objectively & orally to questions asked of him in English/Hindi/Local Language
	Integrity
	The user/individual on the job needs to know and understand how to: SA5. Maintain integrity with respect to company property and time SA6. Communicate with people in a polite manner using respectful language SA7. Resolve difficulties in relationships with colleagues amicably
	Motivation
	The user/individual on the job needs to know and understand how to: SA8. Take responsibility for completing one's own work assignment SA9. Take initiative to enhance/learn skills in one's area of work SA10. Reflect and act upon one's learning from experience
B. Professional Skills	Time Management
	The user/individual on the job needs to know and understand how to: SB1. Prioritize and execute tasks within scheduled time limit
	Reliability
	The user/individual on the job needs to know and understand how to: SB2. Avoid absenteeism and maintain quality of work SB3. Work in a disciplined environment and adhere to working norms of the organisation SB4. Be punctual and adhere to timelines
	Analytical Thinking
	The user/individual on the job needs to know and understand how to: SB5. Suggest methods to avoid accidents/errors while operating mobile equipments SB6. Identify possible ways to improve operational efficiency
	Operations
	The user/individual on the job needs to know and understand how to: SB7. Perform driving operations such as forward, reverse, 'U' turn, tight spot manoeuvre, etc. SB8. Select and use the correct combination of levers to operate the vehicle in different modes.



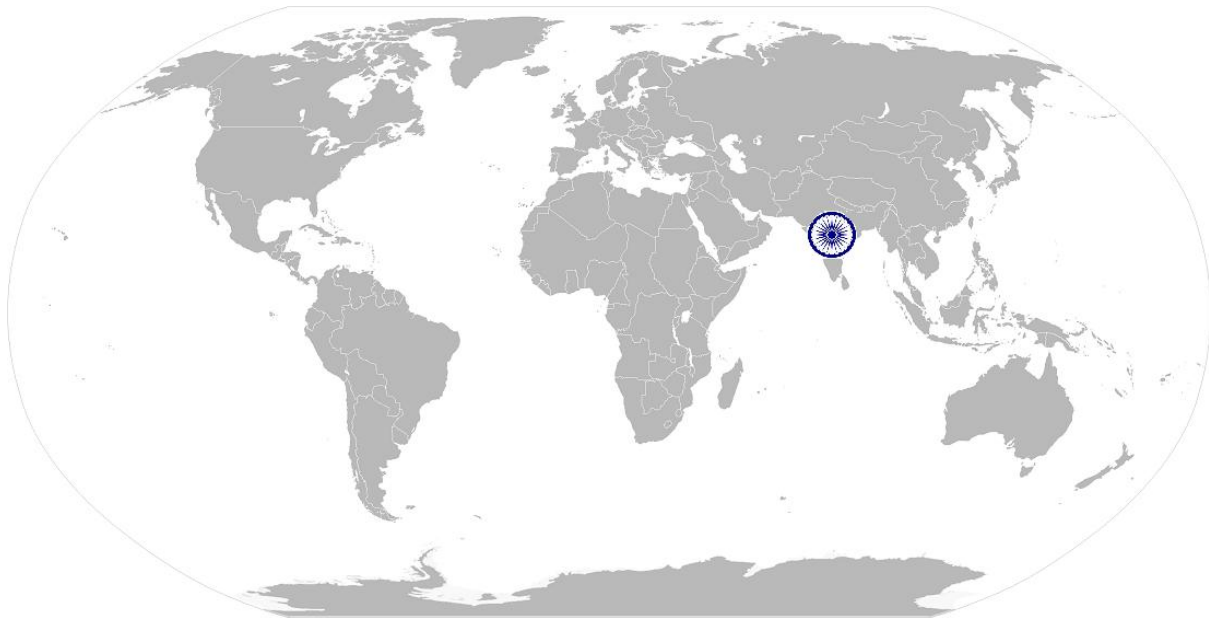
	SB9. Manoeuvre the blade and rear attachments to achieve various tasks
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NOS Version Control

NOS Code	ISC/N0402		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and steel	Drafted on	18/09/2014
Industry Sub-sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory	Last reviewed on	30/12/2014
Occupation	Iron Making	Next review date	30/12/2015





ISC/N0403: Perform basic maintenance and troubleshooting of the mobile equipments



National Occupational Standards

Overview

This unit is about performing routine maintenance and troubleshooting on the Mobile Equipment



Unit Code	ISC/N0403
Unit Title (Task)	Perform basic maintenance and troubleshooting of the mobile equipments
Description	This unit is about performing routine maintenance and troubleshooting tasks on the Vehicle / Mobile Equipment.
Scope	This OS unit/task covers the following: - Carry out routine maintenance in accordance with the manufacturer's recommendations and company procedures - Carry out basic diagnostics and Troubleshooting
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Carry out routine maintenance in accordance with the manufacturer's recommendations and company procedures	To be competent, the user/individual on the job must be able to: PC1. Track machine operating hours to assess the right service schedule. PC2. Clean air filter dust bowls PC3. Clean footplates, pedals and steps free from mud, or dirt. PC4. Drain water and sediment /fuel separators PC5. Replenish coolants, lubricants and fluids PC6. Grease all greasing pins and pivot points PC7. Check battery electrolyte levels and condition of the terminals and carrying out minor adjustments if required. PC8. Check and maintain the tyre rims, air pressure, wheel nuts and treads. PC9. Ensure the machine is on firm and level ground before attempting to carry out any maintenance activity. PC10. Ensure the locking bar is in position to prevent the front and rear chassis moving and creating a crushing zone (articulated machines only) PC11. Ensure that no maintenance task on the engine is performed when the vehicle is running or still hot
Carry out basic diagnostics & Troubleshooting	To be competent, the user/individual on the job must be able to: PC12. Identify any problem with the normal functioning of the forklift PC13. Carry out troubleshooting to ensure completion of given task PC14. Assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/	The user/individual on the job needs to know and understand: KA1. Types of documentation in organization e.g. daily maintenance checklist and importance of the same

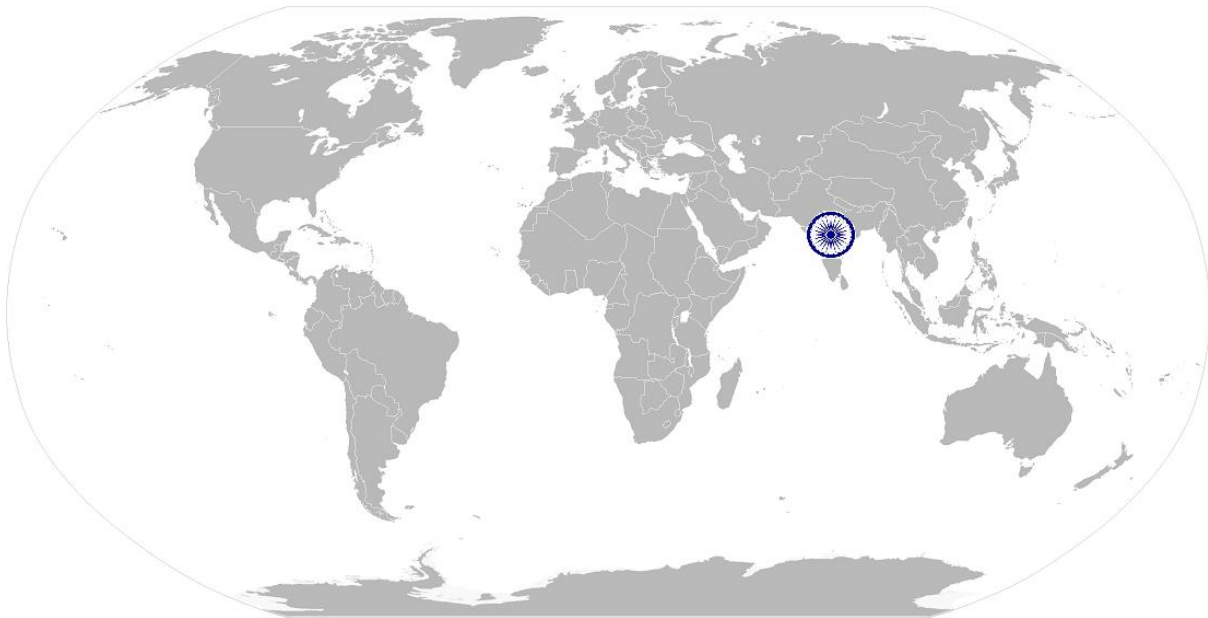


Organisation and its processes)	KA2. Risk and impact of not following defined procedures/work instructions KA3. Rules and regulations of the mill /steel plant as per standard operating procedure (SOP) KA4. Risk and impact of not following company's SOP KA5. Escalation matrix for reporting identified problems
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Controls and switches needed to operate the vehicle properly KB2. Basic mechanism & various functions of the different components of the vehicle & issues of vehicle stability KB3. Technical response to emergencies e.g. fire, explosion etc. KB4. Faults in running engine
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Oral Communication (Listening and Speaking skills) The user/individual on the job needs to know and understand how to: SA1. Communicate with supervisors/juniors/colleagues/casual workers in English/Hindi/Local language (E/H/LL) SA2. Respond objectively & orally to questions asked of him in E/H/LL SA3. Discuss task lists, schedules, and work-loads with the team members
B. Professional Skills	Reliability The user/individual on the job needs to know and understand how to: SB1. Avoid absenteeism and maintain quality of work SB2. Work in a disciplined environment & adhere to working norms of the organisation SB3. Be punctual and adhere to timelines Time Management The user/individual on the job needs to know and understand how to: SB4. Prioritize and execute tasks within the scheduled time limits SB5. Carry out all maintenance work in the given time limit and focus on critical activities first



NOS Version Control

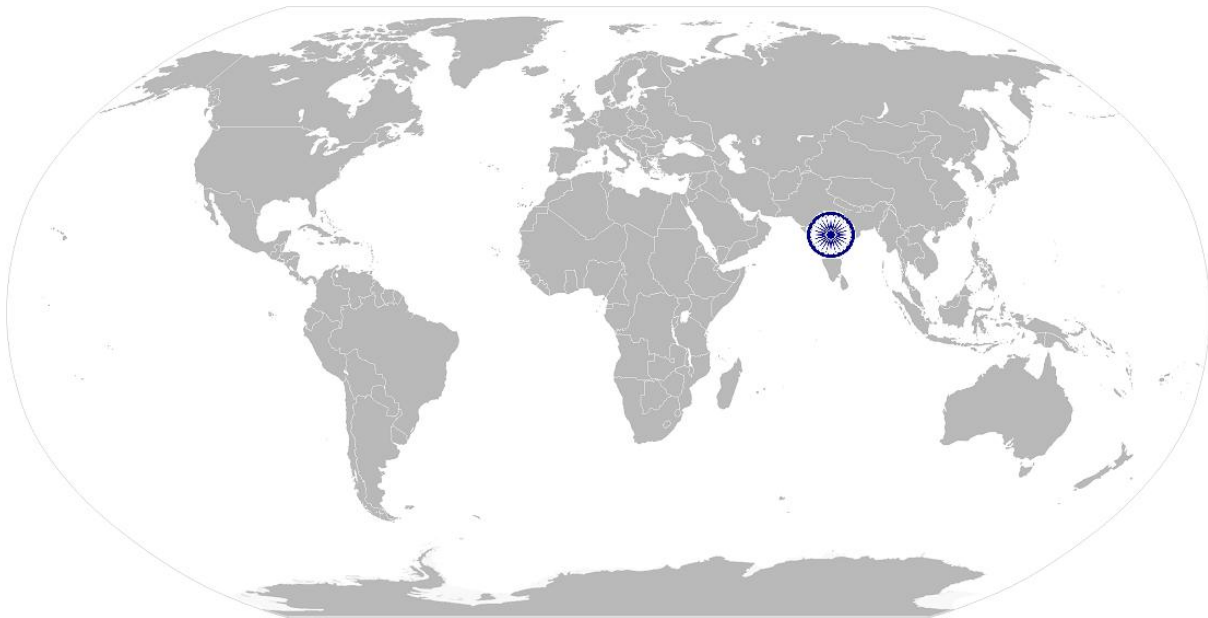
NOS Code	ISC/N0403		
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Industry	Iron and steel	Drafted on	18/09/2014
Industry Sub-sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory	Last reviewed on	30/12/2014
Occupation	Iron Making	Next review date	30/12/2015





ISC/N0008: Use basic health and safety practices at the workplace

National Occupational Standards



Overview

This unit covers health, safety and security at the workplace. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment.



Unit Code	ISC/N0008
Unit Title (Task)	Use basic health and safety practices at the workplace
Description	This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc.
Scope	This unit/task covers the following: • Health and safety procedures • Fire safety procedures • Emergencies, rescue and first aid procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Health and safety procedures	The user/individual on the job should be able to: PC1. Use protective clothing/equipment for specific tasks and work conditions Protective clothing includes: • Leather or asbestos gloves • Flame proof aprons • Flame proof overalls buttoned to neck • Cuff less (without folds) trousers • Reinforced footwear • Helmets/hard hats • Cap and shoulder covers • Ear defenders/plugs • Safety boots • Knee pads • Particle masks • Glasses/gloves/visors Equipment includes: • Hand shields • Machine guards • Residual current devices • Shields • Dust sheets • Respirator PC2. State the name and location of people responsible for health and safety in the workplace

Various areas are listed below:

- On chemical containers
- Equipment
- Packages
- Inside buildings
- Open areas, public places etc.

PC3. State the names and location of documents that refer to health and safety in the workplace

PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace

Hazards include:

- Working with electrical and thermal tools and equipment
- Sharp edged and heavy tools
- Heated metals
- Oxyfuel and gas cylinders
- Welding radiation
- Surfaces: sharp, slippery, uneven, chipped, broken, etc.
- Substances: chemicals, gas, oxy-fuel, fumes, dust, etc.
- Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc.
- Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.

PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role

Safe working practices include:

- Using protective clothing and equipment
- Putting up and reading safety signs
- Handle tools in the correct manner and store and maintain them properly
- Keep work area clear of clutter, spillage and unsafe object lying casually
- While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc.
- Safe lifting and carrying practices
- Use equipment that is working properly and is well maintained
- Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors etc.

Methods are:

- Training in health and safety procedures
- Using health and safety procedures
- Use of equipment and working practices (such as safe carrying procedures)
- Safety notices, advice
- Instruction from colleagues and supervisors

	PC6. State location of general health and safety equipment in the workplace PC7. Inspect for faults, set up and safely use steps and ladders in general use Faults : • Corrosion of metal components • Deterioration • Splits and cracks timber components • Imbalance • Loose rungs • Nuts or bolts, etc. Set up: • Firm/level base • Clip/lash down • Leaning at the correct angle, etc. PC8. Work safely in and around trenches, elevated places and confined areas PC9. Lift heavy objects safely using correct procedures PC10. Apply good housekeeping practices at all times. Good housekeeping practices: • Clean/tidy work areas • Removal/disposal of waste products • Protect surfaces PC11. Identify common hazard signs displayed in various areas PC12. Retrieve and/or point out documents that refer to health and safety in the workplace
Fire safety procedures	The user/individual on the job should be able to: PC13. Use the various appropriate fire extinguishers on different types of fires correctly. Fire extinguishers: • Sand • Water • Foam • Co2 • Dry powder Fires: • Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. • Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances • Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) • Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents) Causes of fires: • Heating of metal

	• Spontaneous ignition • Sparking, • Electrical heating • Loose fires (e.g. Smoking, welding, etc.) • Chemical fires, etc. PC14. Demonstrate rescue techniques applied during fire hazard PC15. Demonstrate good housekeeping in order to prevent fire hazards PC16. Demonstrate the correct use of a fire extinguisher
Emergencies, rescue and first-aid procedures	The user/individual on the job should be able to: PC17. Demonstrate how to free a person from electrocution PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc. PC19. Demonstrate basic techniques of bandaging PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. Few general health and safety equipment are mentioned below : • Fire extinguishers • First aid equipment • Safety instruments and clothing • Safety installations, e.g. Fire exits, exhaust fans etc. PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases PC23. Demonstrate the artificial respiration and the CPR Process PC24. Participate in emergency procedures. Emergency procedures are: • Raising alarm • Safe/efficient evacuation • Correct means of escape • Correct assembly point • Roll call • Correct return to work PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: • Name • Date/time of incident • Date/time of report, • Location • Environment conditions • Persons involved • Sequence of events • Injuries sustained • Damage sustained • Actions taken • Witnesses



	- Supervisor/manager notified Documents: - Fire notices - Accident reports - Safety instructions for equipment and procedures - Company notices and documents - Legal documents (e.g. Government notices) Job titles: - Health and safety officer - First aid officer - Fire officer PC26. Demonstrate correct method to move injured people and others during an emergency
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace KA2. State the names and location of documents that refer to health and safety in the workplace
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB3. Meaning of “hazards” and “risks” KB4. Health and safety hazards commonly present in the work environment and related precautions KB5. Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible KB6. Activities and causes of risk and accident KB7. Methods of accident prevention KB8. Safe working practices when working with tools and machines KB9. Safe working practices while working at various hazardous sites KB10. Where to find all the general health and safety equipment in the workplace KB11. Various dangers associated with the use of electrical equipment KB12. Preventative and remedial actions to be taken in the case of exposure to toxic materials - Exposure: ingested, contact with skin, inhaled - Preventative action: ventilation, masks, protective clothing/equipment - Remedial action: immediate first aid, report to supervisor - Materials: solvents, flux, lead KB13. Importance of using protective clothing/equipment while working KB14. Precautionary activities to prevent the fire accident Activities and causes: - Physical actions - Reading - Listening to and giving instructions

	• Inattention • Sickness and incapacity (e.g. Drunkenness) • Health hazards (e.g. Untreated injuries and contagious illness) KB15. Various causes of fire KB16. Techniques of using the different fire extinguishers KB17. Different methods of extinguishing fire KB18. Rescue techniques applied during a fire hazard KB19. Various types of safety signs and what they mean KB20. Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries KB21. Content of written accident report KB22. Potential injuries and ill health associated with incorrect manual handling KB23. Safe lifting and carrying practices KB24. Personal safety, health and dignity issues relating to the movement of a person by others KB25. Potential impact to a person who is moved incorrectly
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Reading and Writing Skills
	The user/individual on the job needs to know and understand how to:
	SA1. Read and comprehend basic content to read labels, charts, signage's SA2. Read and comprehend basic English to read manuals of operations SA3. Read and write an accident/incident report in local language or English
	Oral Communication (Listening and Speaking skills)
	The user/individual on the job needs to know and understand how to:
B. Professional Skills	SA4. Question co-workers appropriately in order to clarify instructions and other issues SA5. Give clear instructions to co-workers, subordinates others
	Decision Making
	The user/individual on the job needs to know and understand how to:
B. Professional Skills	SA6. Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
	Plan and Organize
B. Professional Skills	The user/individual on the job needs to know and understand:
	SB1. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity

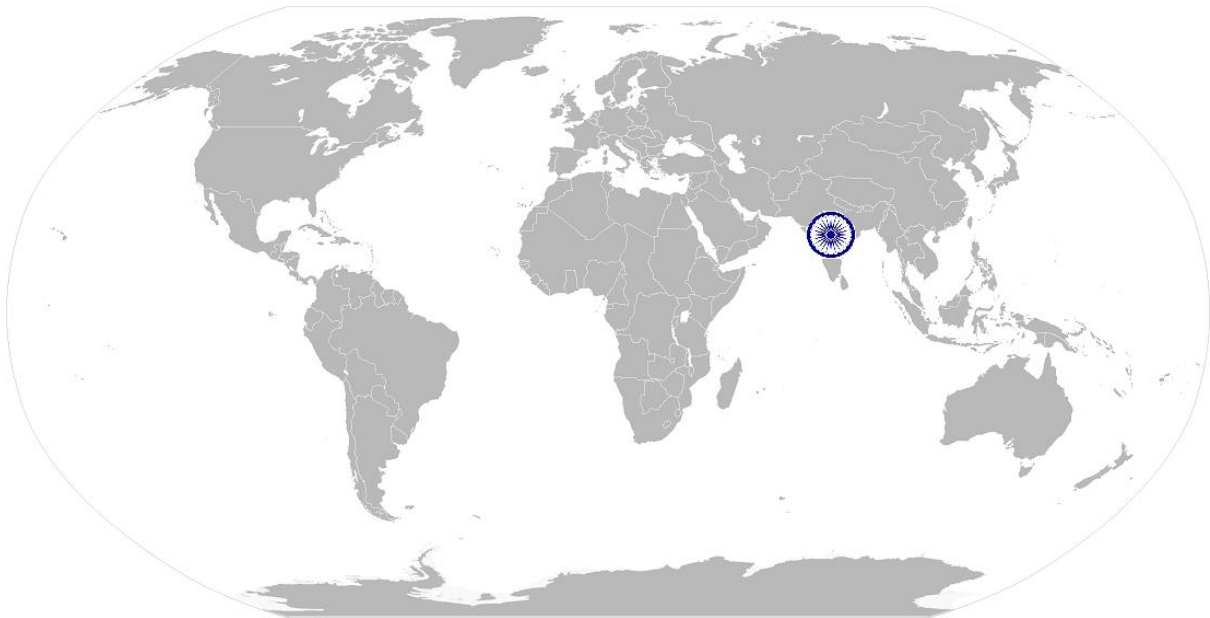


	Working with others
	The user/individual on the job needs to know and understand how to: SB2. Remain congenial while discussing and debating issues with co-workers SB3. Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice SB4. Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives SB5. Thank co-workers for any assistance received SB6. Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
	Problem Solving
	The user/individual on the job needs to know and understand how to: SB7. Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s) SB8. Identify immediate or temporary solutions to resolve delays SB9. Identify sources of support that can be availed of for problem solving for various kind of problems SB10. Seek appropriate assistance from other sources to resolve problems SB11. Report problems that you cannot resolve to appropriate authority
	Analytical Thinking
	The user/individual on the job needs to know and understand how to: SB12. Identify cause and effect relations in their area of work SB13. Use cause and effect relations to anticipate potential problems and their solution



NOS Version Control

NOS Code	ISC/N0008		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and steel	Drafted on	23/07/2014
Industry Sub-sector	All departments	Last reviewed on	30/12/2014
Occupation	Iron Making	Next review date	30/12/2015





ISC/N0009: Works effectively with others

National Occupational Standards



Overview

This unit covers basic practices that improve effectiveness of working with others in an organisational set-up.

Unit Code	ISC/N0009
Unit Title (Task)	Works effectively with others
Description	This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behaviour and interactions with others at the workplace.
Scope	This unit/task covers the following: - Ensure appropriate communication with superiors, peers and others as applicable at work place - Demonstrate appropriate behaviour and etiquette at work place
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Ensure appropriate communication with superiors, peers and others as applicable at work place	The user/individual on the job should be able to: PC1. Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required PC2. Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt PC3. Provide information to others clearly, at a pace and in a manner that helps them to understand
Demonstrate appropriate behaviour and etiquette at work place	The user/individual on the job should be able to: PC4. Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible PC5. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks PC6. Display appropriate communication etiquette while working PC7. Display active listening skills while interacting with others at work PC8. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism PC9. Demonstrate responsible and disciplined behaviours at the workplace PC10. Escalate grievances and problems to
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions KA2. Reporting structure, inter-dependent functions, lines and procedures in the work area KA3. Relevant people and their responsibilities within the work area KA4. Escalation matrix and procedures for reporting work and employment related issues



B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Various categories of people that one is required to communicate and co-ordinate with in the organization KB2. Importance of effective communication in the workplace KB3. Importance of teamwork in organizational and individual success KB4. Various components of effective communication KB5. Key elements of active listening KB6. Value and importance of active listening and assertive communication KB7. Barriers to effective communication KB8. Importance of tone and pitch in effective communication KB9. Importance of avoiding casual expletives and unpleasant terms while communicating professional circles KB10. How poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer KB11. Importance of ethics for professional success KB12. Importance of discipline for professional success KB13. What constitutes disciplined behaviour for a working professional KB14. Common reasons for interpersonal conflict KB15. Importance of developing effective working relationships for professional success KB16. Expressing and addressing grievances appropriately and effectively KB17. Importance and ways of managing interpersonal conflict effectively
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Reading and Writing Skills
	The user/individual on the job needs to know and understand how to: SA1. Read and comprehend basic content to read labels, charts, signage's SA2. Read and comprehend basic English to read manuals of operations SA3. Read and write an accident/incident report in local language or English
	Oral Communication (Listening and Speaking skills)
	The user/individual on the job needs to know and understand how to: SA4. Question co-workers appropriately in order to clarify instructions and other issues SA5. Provide clear instructions to co-workers, subordinates others
	Decision Making
	The user/individual on the job needs to know and understand how to: SA6. Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down

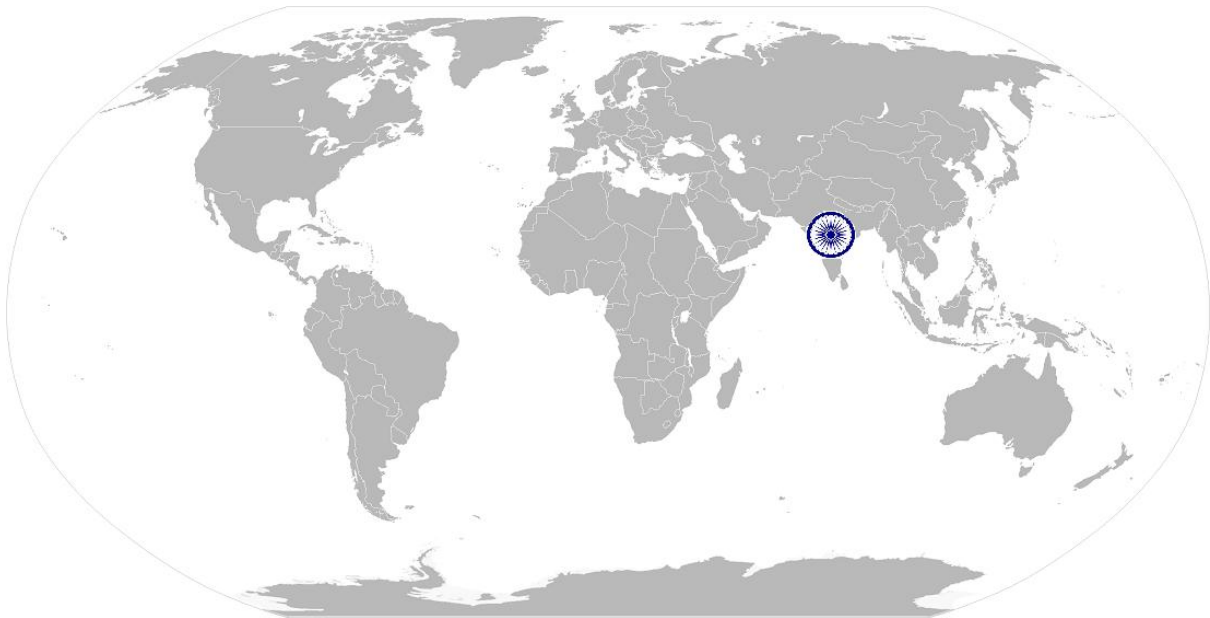


	procedure and guidelines
B. Professional Skills	Plan and Organize
	The user/individual on the job needs to know and understand:
	SB1. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
	Working with others
	The user/individual on the job needs to know and understand how to:
	SB2. Remain congenial while discussing and debating issues with co-workers SB3. Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice SB4. Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives SB5. Thank co-workers for any assistance received SB6. Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
	Problem Solving
	The user/individual on the job needs to know and understand how to:
	SB7. Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s) SB8. Identify immediate or temporary solutions to resolve delays SB9. Identify sources of support that can be availed of for problem solving for various kind of problems SB10. Seek appropriate assistance from other sources to resolve problems SB11. Report problems that you cannot resolve to appropriate authority
	Analytical Thinking
	The user/individual on the job needs to know and understand how to:
	SB12. Identify cause and effect relations in their area of work SB13. Use cause and effect relations to anticipate potential problems and their solution



NOS Version Control

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Credits(NSQF)	TBD	Version number	1.0
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CRITERIA FOR ASSESSMENT OF TRAINEES

Job Role: Mobile Equipment Operator

Qualification Pack: ISC/Q0401

Sector Skill Council: Indian Iron & Steel Sector Skill Council

Guidelines for Assessment:

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below)
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criteria.
5. To pass the Qualification Pack, every trainee should score a minimum of 60% in every NOS.
6. In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

NOSs	PCs	Marks Allocated			
		Total Marks 1000	Out Of	Theory	Practical
ISC/N0401: Prepare the vehicle	PC1. Interpret and understand the job requirements	200	25	10	15
	PC2. Prepare a work plan to carry out the job		15	5	10
	PC3. Check oil levels of engine, transmission, radiator coolant and brake		15	0	15
	PC4. Check differential and hydraulic oil levels		15	5	10
	PC5. Check the hydraulic hose and ram for leakages		10	0	10
	PC6. Check condition of parking brake, main horn, reverse horn, and head light		10	0	10
	PC7. Check the various controls, gauges, warning lamp and other safety devices		20	5	15
	PC8. Check fan belt tension, electrolyte level and terminal tightness		15	5	10



	PC9. Check and adjust driving position, rear and side mirrors, seat belts		5	0	5
	PC10. Clean air filter dust bowls		10	5	5
	PC11. Drain water and sediment from the fuel tank		5	0	5
	PC12. Top up coolant and oil in engine, transmission, etc. if necessary		10	0	10
	PC13. Check electrolyte level, terminals, seating arrangements of the battery		5	0	5
	PC14. Apply grease to all greasing pins and pivot points		5	0	5
	PC15. Ensure the locking bar is in position to prevent the front and rear chassis moving and creating a crushing zone (articulated machines only)		15	5	10
	PC16. Keep footplates and steps clean and free from mud, dirt and oil.		5	0	5
	PC17. Report defects precisely to the supervisor if beyond scope of his role		15	5	10
		Total	200	45	155
ISC/N0402: Perform mobile equipment operations	PC1. Start the engine using the push-button switch and key switch together	400	25	5	20
	PC2. Use the priming pump and pre-heater to start the engine in cold weather conditions		20	10	10
	PC3. Load the material to be transferred safely & correctly, ensuring that it is not damaged or spilt or does not slip/fall off during loading or during journey		40	10	30
	PC4. Always wear safety belt & helmet while driving		10	0	10
	PC5. Ensure proper functioning of rear view mirror and audio visual alarm		10	0	10
	PC6. Not use mobile phones while driving		10	0	10
	PC7. Look out for pedestrians, people working, stationary objects and hazards such as trenches, potholes, cables, caring for overhead pipes, power lines, confined space etc.		25	0	25



	PC8. Operate the shifting quadrant device for selecting the correct gear setting		35	10	25
	PC9. Ensure vehicle load and operating speed are within limits specified by the manufacturer & do not cross the limits specified in the company		40	10	30
	PC10. Adhere to time limits given by supervisor		20	5	15
	PC11. Follow the road signs and locally developed signals		30	10	20
	PC12. Keep a safe distance from a tip edge and use an approved stop block before tipping over an edge		15	5	10
	PC13. Select & use the right type of brake in different situations during journey		20	5	15
	PC14. Discharge the load safely at the position and in the manner designated by the supervisor		30	5	25
	PC 15. Ensure that vehicle is always parked on firm, level ground; with handbrake applied and drive & controls disengaged		25	0	25
	PC16. Put the front accessory (boom, fork, basket/bucket) in lowered condition touching the ground; switch the engine off, apply handbrake and alight from the seat		30	5	25
	PC17. Ensure that nobody except the authorised driver ever boards, travels on or stands near the Vehicle		15	5	10
		Total	400	85	315
ISC/N0403: Perform basic maintenance and troubleshooting of the mobile equipments	PC1. Track machine operating hours to assess the right service schedule.	150	25	10	15
	PC2. Clean air filter dust bowls		5	0	5
	PC3. Clean footplates, pedals and steps free from mud, or dirt.		5	0	5
	PC4. Drain water and sediment /fuel separators		5	0	5
	PC5. Replenish coolants, lubricants and fluids		10	0	10



	PC6. Grease all greasing pins and pivot points		5	0	5
	PC7. Check battery electrolyte levels and condition of the terminals and carrying out minor adjustments if required.		10	0	10
	PC8. Check and maintain the tyre rims, air pressure, wheel nuts and treads.		5	0	5
	PC9. Ensure the machine is on firm and level ground before attempting to carry out any maintenance activity.		5	0	5
	PC10. Ensure the locking bar is in position to prevent the front and rear chassis moving and creating a crushing zone (articulated machines only)		15	5	10
	PC11. Ensure that no maintenance task on the engine is performed when the vehicle is running or still hot		10	0	10
	PC12. Identify any problem with the normal functioning of the forklift		15	5	10
	PC13. Carry out troubleshooting to ensure completion of given task		20	5	15
	PC14. Assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel		15	5	10
		Total	150	30	120
ISC/N0008: Use basic health and safety practices at the workplace	PC1. Use protective clothing/equipment for specific tasks and work conditions	150	10	5	5
	PC2. State the name and location of people responsible for health and safety in the workplace		5	0	5
	PC3. State the names and location of documents that refer to health and safety in the workplace		1	0	1
	PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace		9	5	4



PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role	10	5	5
PC6. State location of general health and safety equipment in the workplace	5	0	5
PC7. Inspect for faults, set up and safely use steps and ladders in general use	5	0	5
PC8. Work safely in and around trenches, elevated places and confined areas	5	0	5
PC9. Lift heavy objects safely using correct procedures	5	0	5
PC10. Apply good housekeeping practices at all times	5	0	5
PC11. Identify common hazard signs displayed in various areas	6	5	1
PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	4	0	4
PC13. Use the various appropriate fire extinguishers on different types of fires correctly	10	5	5
PC14. Demonstrate rescue techniques applied during fire hazard	10	5	5
PC15. Demonstrate good housekeeping in order to prevent fire hazards	5	0	5
PC16. Demonstrate the correct use of a fire extinguisher	5	0	5
PC17. Demonstrate how to free a person from electrocution	5	0	5
PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	5	0	5
PC19. Demonstrate basic techniques of bandaging	5	0	5
PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	10	5	5



	PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments		5	0	5
	PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases		5	0	5
	PC23. Demonstrate the artificial respiration and the CPR Process		4	0	4
	PC24. Participate in emergency procedures		4	0	4
	PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible		6	5	1
	PC26. Demonstrate correct method to move injured people and others during an emergency		1	0	1
		Total	150	40	110
ISC/N0009: Works effectively with others	PC1. Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	100	10	5	5
	PC2. Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt		10	5	5
	PC3. Provide information to others clearly, at a pace and in a manner that helps them to understand		10	0	10
	PC4. Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible		10	5	5
	PC5. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks		10	5	5
	PC6. Display appropriate communication etiquette while working		10	0	10



	PC7. Display active listening skills while interacting with others at work		10	0	10
	PC8. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism		10	5	5
	PC9. Demonstrate responsible and disciplined behaviours at the workplace		15	5	10
	PC10. Escalate grievances and problems to supervisor		5	0	5
		Total	100	30	70

